UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2011 question paper for the guidance of teachers

5070 CHEMISTRY

5070/41 Paper 4 (Alternative to Practical), maximum raw mark 60

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1 26 (1) cm^3 [1]

2 **(a)** red to blue (1)

(b) **(i)** hydrogen (1) pops in a flame (1)

(c) **(i)** effervescence or fizzing or bubbles given off (1)
Not gas evolved

(ii) carbon dioxide (1) turns lime water milky or white (1)
ecf on O_2 [6]

3 **(a)** 0.48 (1) g

(b) **(i)** silver/grey/shiny metal/solid (1)

(ii) white solid/powder (1)

(c) to ensure constant weight or that reaction was complete (1)

(d) **(i)** 0.8 (1) g

(ii) 0.32 (1) g

(e) 0.48/24 = 0.02 0.32/16 = 0.02 (1)
MgO (1)

(f) **(i)** $MgO + 2HCl \rightarrow MgCl_2 + H_2O$ (1)
or $+ H_2SO_4 + HNO_3$

(ii) basic (1) [10]

4 (c) (1) [1]

5 (b) (1) [1]

6 (d) (1) [1]

7 (d) (1) [1]

8 (b) (1) [1]

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9 **(a)** 1.22 (1) g

(b) to allow gas/carbon dioxide to escape (1)

(c) red/pink to yellow (1)

(d)

24.1	41.1	28.5	1 mark for each correct row or column (3)
0.0	17.6	4.8	
24.1	23.5	23.7	

Mean value = 23.6 (1) cm^3

(e) 0.00236 (1)

(f) 0.00236 (1)

(g) 0.0236 (1)

(h) 0.05 (1)

(i) 0.0264 (1)

(j) $MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$ (1)

(k) 0.0132 (1)

(l) **(i)** 84 (1)

(ii) 1.11 (1) g

(iii) 1.11/1.22 = 91% (1) [17]

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10 **(a)** transition metal or transition metal ions present (1)

(b) **(i)** blue ppt (1)

(ii) insoluble in excess (1)

(c) **(i)** blue ppt (1)

(ii) soluble forming a DEEP blue solution (1)

(d) $HNO_3/AgNO_3$ (2)
White ppt (1)
$CuCl_2$ (1) [9]

11 **(a)** 26.8, 28.5, 30.3, 31.2 (1) all correct
1.8, 3.5, 5.3, 6.2 (1) all correct

(b) all points plotted correctly (1)
two intersecting straight lines, the first of which must pass through zero (2).
points joined by a curve or a series of straight lines at intersection (1)

(c) **(i)** 0.34 (1) g

(ii) 0.70 (1) g

(iii) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (1)

(iv) redox or displacement or exothermic (1)

(v) $50 \times conc^n / 1000 = 0.70/56$ (1)
$Conc^n = 0.25$ (1) mol/dm^3

(d) blue colour disappears
or red deposit/solid/copper at bottom of beaker (1) [12]

[with all graphical answers please read candidate's graph and to accuracy of ± half small square]

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